

Adding age effects

a4a Training Course — Module 2.3

Danai Mantopoulou Palouka & Alessandro Ligas

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The signal we are answering

The mean model left a clear **age** pattern in both data sources.

The method

Explore the age effect on F and on q **in isolation**, then bring them together.
Change one thing, refit, look at the residuals again.

The procedure is carried by the residuals: in a well specified model they should show no trend and no large outliers.

An age effect on F

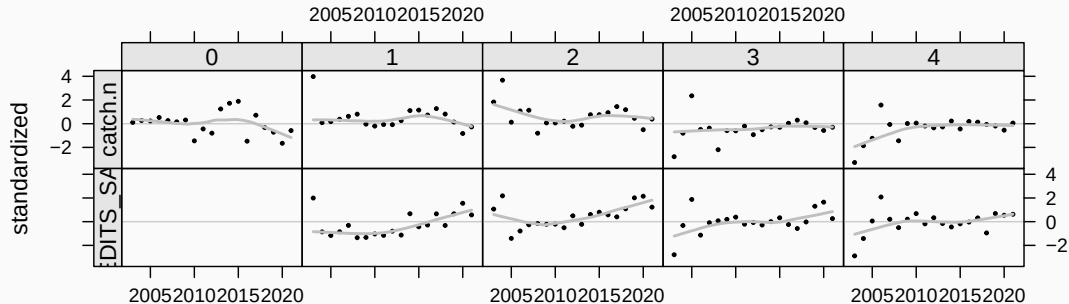
```
fit02 <- sca(stk, idx,  
            fmodel = ~factor(age),    # <- was ~1  
            qmodel = list(~1),  
            srmodel = ~1, vmmodel = list(~1, ~1), n1model = ~1)
```

`factor(age)` means **as many parameters as ages**, each one independent of the others.

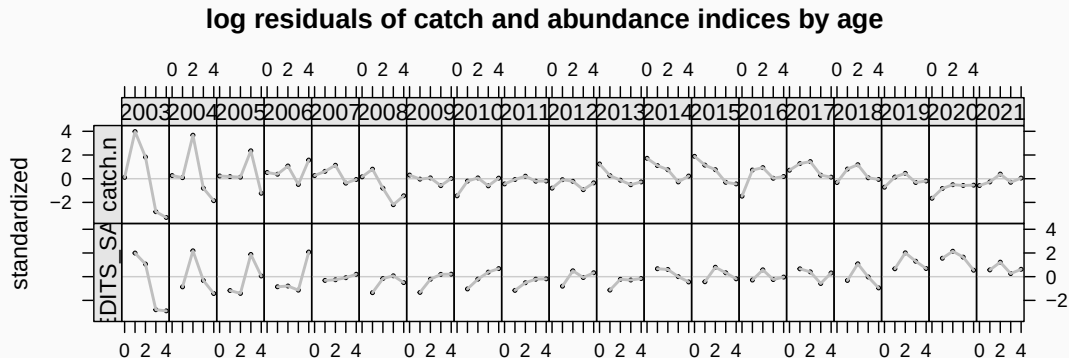
The catch-at-age residuals become less staggered, reflecting the age effect. The year trends are still there, but the average residual **by age** is now much closer to 0.

Age effect on F — residuals by year

log residuals of catch and abundance indices by year



Age effect on F — residuals by age



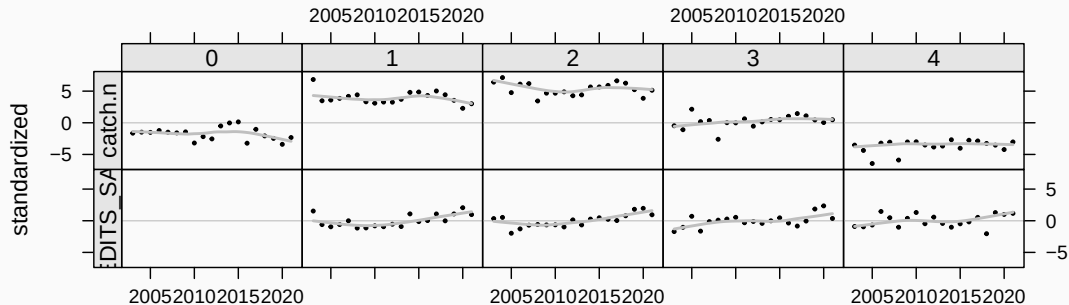
An age effect on q instead

Now add the age effect to the catchability submodel, and **remove** it from the catch.

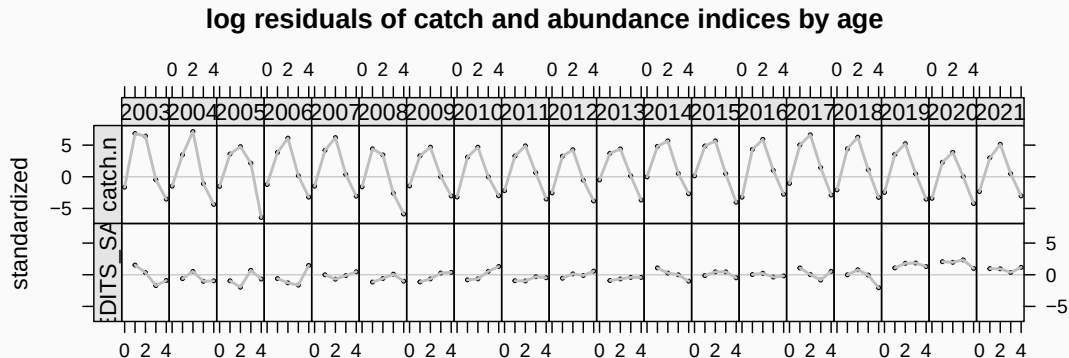
```
fit03 <- sca(stk, idx,  
            fmodel = ~1,  
            qmodel = list(~factor(age)), # <- was ~1  
            srmodel = ~1, vmmodel = list(~1, ~1), n1model = ~1)
```

Age effect on q — residuals by year

log residuals of catch and abundance indices by year



Age effect on q — residuals by age

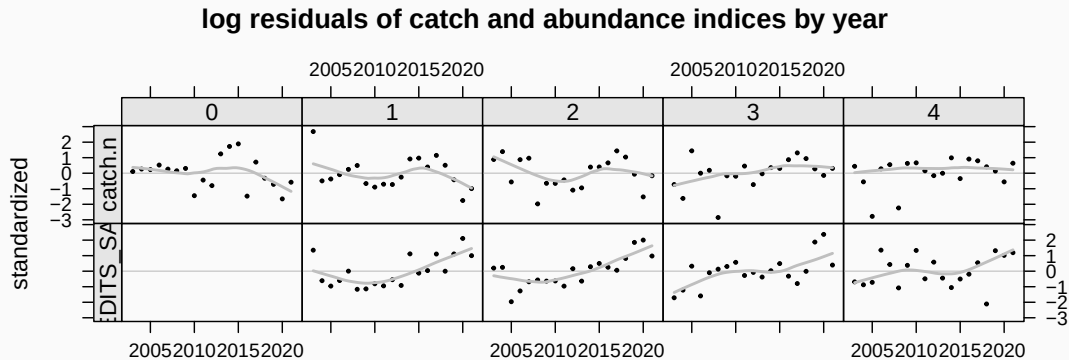


Both together

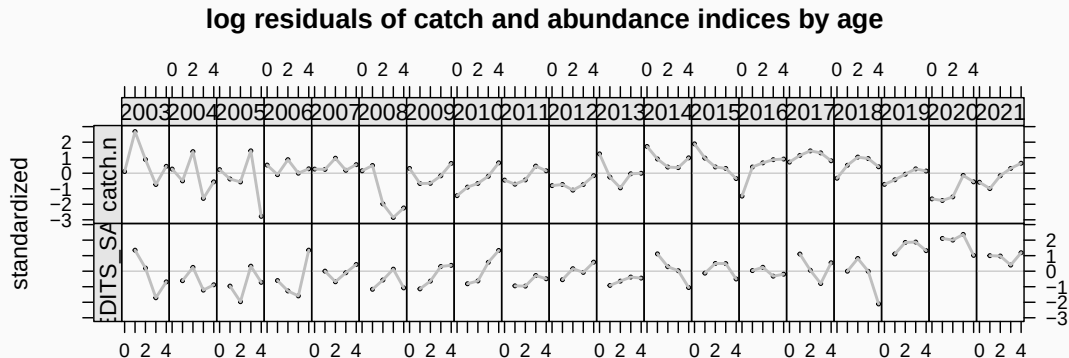
```
fit04 <- sca(stk, idx,  
            fmodel = ~factor(age), qmodel = list(~factor(age)),  
            srmodel = ~1, vmodel = list(~1, ~1), n1model = ~1)
```

Finally the two effects are brought together.

Both effects — residuals by year



Both effects — residuals by age



Now look at what it estimated

fbar	ssb
4.874673e-15	3.240313e+17

 Converged, clean residuals, and nonsense

$\bar{F} \approx 5 \times 10^{-15}$ and $SSB \approx 3 \times 10^{17}$ tonnes: no fishing at all, on an impossibly heavy stock.

Residuals tell you whether the model reproduces the **data**. They cannot tell you whether the answer is **possible**. Always look at $\bar{F}$, SSB and recruitment as well.

Adding a year effect to F — next module — brings $\bar{F}$ back to 0.6–2.8.

Summary

- Explore the age effect on F and on q **separately**, then together
- `factor(age)` = one free parameter per age, independent of its neighbours
- On F : the catch residuals are less staggered and centre on 0 by age
- The year trends remain — that is Module 2.4
- **Check $\bar{F}$ and SSB before you believe any residual plot**

Exercise

Open `handson_M2.3_age_effects.R`.

1. Fit `fit02`, `fit03` and `fit04`
2. Plot the residuals of each **by year and by age**
3. Compare the fitted stocks with `FLStocks()`